Technical Documentation

ACDT Project 1, GROUP32: Is daily social media use associated with student mental health?
Website: **[GitHub Pages URL]** Code repository: **[GitHub repository URL]**

1. Research question and hypotheses

Outcome Y = Mental_Health_Score; treatment X = Avg_Daily_Usage_Hours. Model: $Y = \beta_0 + \beta_1 X + \varepsilon$. H_0: $\beta_1 = 0$; H_1: $\beta_1 \neq 0$ (two-sided; our earlier directional version was $\beta_1 < 0$, controlling for academic level). **$\alpha = 0.05$**: a Type I error would justify blanket screen-time rules and wasted counselling time; a Type II error would leave struggling heavy users unflagged. Neither cost dominates, so we kept the conventional level.

2. Data and preprocessing

Source: *Students' Social Media Addiction* (Kaggle), 705 rows, 13 self-reported variables, no missing values. Recruitment method and sampling frame are not documented (treated as a convenience sample). Student IDs 242 and 274 are identical in every answer; one was removed (**n = 704**). 110 countries, median 1 student per country. For DML, Country was recoded to the 10 largest countries plus "Other".

Variable	Type / range	Role	Reason
Avg_Daily_Usage_Hours	Continuous, 1.5–8.5 h	Treatment (X)	The conventional wisdom is about time spent
Mental_Health_Score	Integer, observed 4–9 (1–10 scale)	Outcome (Y)	Replaces GPA, which is not in the data
Age, Gender, Academic_Level, Relationship_Status	Background	Control	Fixed before usage; can confound, not caused by usage
Country	110 categories	Control / cluster	Strongest confounder; breaks independence
Most_Used_Platform	12 categories	Moderator	Proxy for type of use (our original idea)
Sleep_Hours_Per_Night	Continuous	Mediator	Usage can reduce sleep; not controlled for the total effect
Addicted_Score, Conflicts_Over_Social_Media	Scores / counts	Not controlled	Likely consequences of usage (bad controls)
Affects_Academic_Performance	Yes / No	Secondary outcome	Closest proxy for the original GPA question

3. Methods

Framework level	Method	Implementation details
Sampling distribution / Inference	OLS, t-test, 95% CI	df = 702, $t^* = 1.963$. Independence violated by country clustering, so CR1 cluster-robust SEs with 110 clusters are also reported. Residual normality: Shapiro–Wilk $p < 0.001$; with n = 704 the CLT supports t-based inference.
Prediction	Train/test validation	200 random 80/20 splits (scikit-learn, seeds 0–199); metrics: test R^2, RMSE, vs a train-mean baseline. Prediction intervals: $\hat{y} \pm t^* \cdot s\sqrt{(1 + 1/n + (x-\bar{x})^2/S_{xx})}$.
Inference (heterogeneity)	Per-platform OLS; nested F-test	Bootstrap CIs (2,000 resamples) for slope and R^2. F-test: platform-specific slopes vs one common slope, both with platform intercepts (platforms with $n \geq 10$ and varying outcome).
Reasoning	Country fixed effects; mediation	110 country dummies. Sleep mediation by product of coefficients a·b, 1,000 bootstrap resamples.
Reasoning	Double/debiased ML (partially linear model)	$Y = \theta X + g(W) + \varepsilon$, $X = m(W) + v$. Random forests (100 trees, min leaf 5; 8 in subgroups) estimate g and m with 5-fold cross-fitting; θ from residual-on-residual regression; 3 repetitions, median aggregation; influence-function SE. Covariate sets A–D below.

4. Results

Model	Estimate per hour	95% CI	R^2
OLS: hours only	−0.704	−0.74 to −0.66 (clustered: −0.80 to −0.61)	0.642
OLS: + academic level	−0.691	−0.73 to −0.65	0.657
OLS: + sleep (direct effect)	−0.566	−0.63 to −0.50	0.657
OLS: + country fixed effects	−0.575	−0.62 to −0.53	0.848
OLS: + country + platform	−0.521	−0.57 to −0.47	0.874
OLS: + sleep + addiction + conflicts	−0.063	−0.11 to −0.02	0.896
DML A: background (age, gender, level, relationship, country)	−0.633	−0.68 to −0.59	–
DML B: A + platform	−0.589	−0.65 to −0.53	–
DML C: A + sleep	−0.454	−0.52 to −0.38	–
DML D: A + addiction + conflicts	0.056	0.01 to 0.10	–

Validation: mean test R^2 0.635 (range 0.484–0.730), RMSE 0.662 vs baseline 1.104. Platform slope heterogeneity: $F(8, 650) = 3.59$, $p = 0.0004$; academic level $p = 0.81$; gender $p = 0.31$. Mediation: $a = -0.709$, $b = 0.194$, indirect = −0.138 (95% CI −0.177 to −0.104), 19.6% of total. DML subgroups: Undergraduate −0.63, Graduate −0.68, High School −0.70, Female −0.63, Male −0.69, Instagram −0.66, TikTok −0.54, Facebook −0.66; WhatsApp could not be estimated because scores are constant within its countries. Secondary outcome: students reporting academic harm use 5.54 vs 3.80 h (Welch $t = 25.4$, $p < 0.001$).

Interpretation of D: controlling for addiction and conflicts removes the effect because both are probably downstream of usage and strongly tied to the outcome ($r = -0.95$ for addiction). We report D only to show why such variables must not be controlled.

5. Limitations and threats to validity

- **External validity:** undocumented, non-random sample; half of the 110 countries contribute one student.
- **Internal validity:** cross-sectional self-reports; reverse causation and unmeasured confounding (stress, workload, purpose of use) remain. DML adjusts only for measured covariates; no valid instrument was available.
- **Data quality:** scores identical for all users of LINE, KakaoTalk and VKontakte and within 13% of countries with $n \geq 2$; academic-impact answers move from 0% to 100% across usage bands; outcome is ordinal with six observed values.
- **Model assumptions:** non-normal residuals and clustered observations; addressed with large-sample inference and cluster-robust SEs, not eliminated.

6. Reproducibility

Repository files: **analysis.py** (all computations, writes results.json), **results.json**, **data/Students_Social_Media_Addiction.csv**, **index.html** (data story website), **tools/** (page and document builders). Environment: Python 3 with pandas, numpy, scipy, scikit-learn. Run **python analysis.py** (about 1–2 minutes), then **python tools/build_page.py** and **python tools/build_docs.py**. All random procedures use fixed seeds.

AI assistance: [Team to complete: tools used, for which parts, and how outputs were verified, following the Intercollege AI policy.]